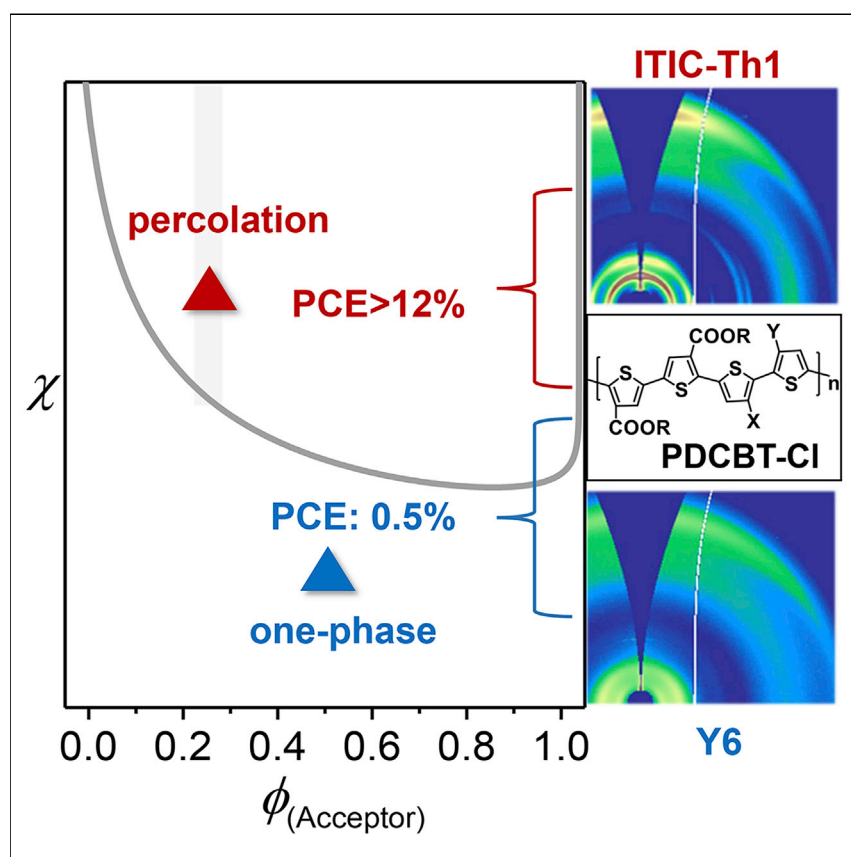


Article

Optimization Requirements of Efficient Polythiophene:Nonfullerene Organic Solar Cells



This research underscores the importance of considering both miscibility and molecular ordering in the design of higher efficiency polythiophene (PT):nonfullerene solar cells. We find that ITIC-Th1 exhibits proper miscibility and relatively highly ordered molecular packing with PDCBT-CI, affording the record performance of >12%. Conversely, due to the excessively high miscibility of PDCBT-CI and Y6, the system exhibits abysmal photovoltaic efficiency of ~0.5%. The results provide efficient guidelines for materials matching of higher efficiency PT:nonfullerene solar cells.

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HIGHLIGHTS

Full structural characterizations of the vitally important PT:nonfullerene systems

New design rules to rationally screen high-efficiency PT:nonfullerene systems

The first-ever report of the spectacular failure of the star nonfullerene acceptor Y6

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Article

Optimization Requirements of Efficient Polythiophene:Nonfullerene Organic Solar Cells

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SUMMARY

Polythiophene (PT) and its derivatives have attracted long-standing attention in the organic photovoltaic community for their low cost and high scalability of synthesis. However, due to the lack of rational guidelines in controlling morphology and matching materials, the power conversion efficiencies (PCEs) based on PTs reported so far are generally below 10%. Here, we establish the first-ever relationship between miscibility, morphology, and device performance of binary blends, based on various nonfullerene acceptors (ITIC-Th1, ITIC, IT4F, IDIC, and Y6) and a PT derivative named PDCBT-Cl by scattering and calorimetric characterizations. Benefiting from a properly quenched mixed phase, PDCBT-Cl:ITIC-Th1 system shows the best efficiency of over 12%. Conversely, the blend of PDCBT-Cl and the star acceptor Y6 remained in a homogeneous state due to their high miscibility, resulting in abysmal performance with PCE of 0.5%. Specific guidelines are also proposed to remediate the performance of PDCBT-Cl:Y6, which are crucial for advancing their practical applications.

INTRODUCTION

With the discovery of a series of novel nonfullerene acceptors,¹ the PCEs of OPVs have been recently boosted significantly, with PCEs up to 17%~18%.^{2–4} A majority of these high-efficiency OPVs are based on the push-pull type conjugated polymer donors. For instance, a widely used push-pull donor named PM6, can realize efficiencies of over 17% by blending with narrow band-gap acceptors.^{3–5} Each of the two different monomers of these donor polymers usually needs several synthesis steps with relatively low total yields, and therefore, push-pull polymers always need more synthesis steps and are costly (>1,000 \$/g) and can seldom be produced in a high quantity up to ~10 g.⁶ The high cost and poor scalability significantly limit the applications of push-pull polymers in OPV commercialization.

In contrast to the aforementioned push-pull polymers, PT and its derivatives are one of the most easily accessible and mass-producible candidates in terms of cost and commercial applications.^{7–10} In recent years, several groups have started to overcome the inferior performance limitations of OPVs, based on PT and nonfullerene acceptors.^{9,11,12} By manipulating the film-forming kinetics, Han group successfully boosted the PCE to 7.2% for P3HT:O-IDTBR system by adding a high boiling point solvent, 1,2,4-trichlorobenzene (TCB). The addition of TCB adjusted the order of crystallization for polymer and acceptor and formed a highly crystalline nano-interpenetrating network.¹² Peng et al. further promoted the PCE of P3HT-based nonfullerene OPVs to an impressive value (8.25%) by changing the aging time of active

Context & Scale

Recently, the efficiencies of organic photovoltaics (OPVs) have surpassed 18%. The realization of these high-efficiency OPVs is based on the use of push-pull type polymer donors, which are costly and can seldom be produced in high quantity. In contrast, polythiophenes (PTs) hold great promises in cost and scalability, rendering them alternatives to the push-pull type donors for commercial applications. Here, we reveal the crucial role of miscibility and molecular ordering in determining the performance of PT:nonfullerene systems. Our study underscores the need for nonfullerene acceptors to match PTs thermodynamically and provides design rules for higher efficiency PT:nonfullerene solar cells. We also emphasize the effectiveness of using the physical parameters to exclude low-performance pairs and screen promising PT:nonfullerene pairs. This approach will accelerate the development of low-cost yet high-efficiency PT:nonfullerene solar cells and be applicable to more OPV systems.

layer solution to adjust the preaggregation state and eventually attained a more favorable morphology.¹¹ As the highest occupied molecular orbital (HOMO) of P3HT is high-lying (~ 5.0 eV), the PCEs of P3HT:nonfullerene OPVs are all less than 10%, which is primarily limited by their relatively low open-circuit voltages (V_{oc}) of <0.9 V.¹³ Thus, in 2016, Hou and co-workers were able to break the 10% efficiency in PT:nonfullerene devices by pairing a carboxylate-substituted PT named PDCBT, with ITIC and a high V_{oc} of 0.94 V was achieved.¹⁴ Inspired by the success of PDCBT, our group further incorporated chlorine atoms into the backbone of PDCBT and synthesized a series of chlorinated polymers. One notable example of these chlorinated polymers is PDCBT-Cl. The monomers used to make PDCBT-Cl can be synthesized on a relatively large scale from commercially available chemicals with high reaction yields and few steps.¹⁵ Currently, high performance nonfullerene OPVs based on PTs afford relatively poor PCEs of 9%~11%,^{11,14–16} which lag significantly behind devices employing PM6-type polymer donors.^{17–20} This is primarily due to the difficulty to screen a suitable acceptor out of thousands of nonfullerene acceptors and form an optimized morphology with PTs. Despite the increasing interest in PT:nonfullerene OPV devices,^{16,21} little attention has been placed on identifying the key thermodynamic, casting kinetic, and morphological factors that control the performance of PT:nonfullerene based blends. Thereby, the fundamental morphology research and understanding of PT based systems are profoundly desired and a guiding principle is extremely important for further improving the device performance and stability of these potentially low-cost systems.

To identify key affecting factors and explore the materials matching rules in PT:nonfullerene OPVs, we characterized the donor:acceptor miscibility, active layer morphology, and relevant device parameters of the solar cells based on PDCBT-Cl and five different nonfullerene acceptors, including ITIC-Th1, ITIC, IT4F, IDIC, and Y6 (Figure 1). These selected acceptors are classified into two sets: (1) variation of end groups: ITIC-Th1, ITIC, and IT4F; (2) variation of central units: ITIC-Th1, IDIC, and Y6. The thermodynamic amorphous-amorphous interaction parameters (χ_{aa}) of these five blends were estimated by differential scanning calorimetry (DSC) and contact angle measurements. We also applied DSC to derive the crystalline-amorphous interaction parameters (χ_{ca}) of PT:nonfullerene OPVs based on a self-consistent model of crystalline-amorphous blends. The crystallinity and phase separation of these blends films were investigated by DSC and synchrotron-based techniques, e.g., grazing incidence wide-angle X-ray scattering (GIWAXS) and resonant soft X-ray scattering (RSOXS). The results allowed us to establish miscibility-morphology-function relations in PT:nonfullerene OPVs. Among the five acceptors, ITIC-Th1 exhibited proper miscibility and relatively highly ordered molecular packing with PDCBT-Cl, affording the best performance of $>12\%$ efficiency with the highest short-circuit current density (J_{sc}) and fill factor (FF). In sharp contrast, due to the excessively high miscibility (hyper-miscibility) of PDCBT-Cl and Y6, the system exhibited abysmal performance with PCE of $\sim 0.5\%$ and stable morphology. Furthermore, this comparative study underscores the importance of considering both miscibility and the ordering of molecular packing in the design of higher efficiency PT:nonfullerene solar cells and proposes practical solutions.

RESULTS AND DISCUSSION

Solar Cell Performance Of PT:Nonfullerene Blends

First, to compare the solar cell performance of PDCBT-Cl:nonfullerene blends, we fabricated OPV devices with PDCBT-Cl as donor and ITIC, ITIC-Th1, IT4F, IDIC,

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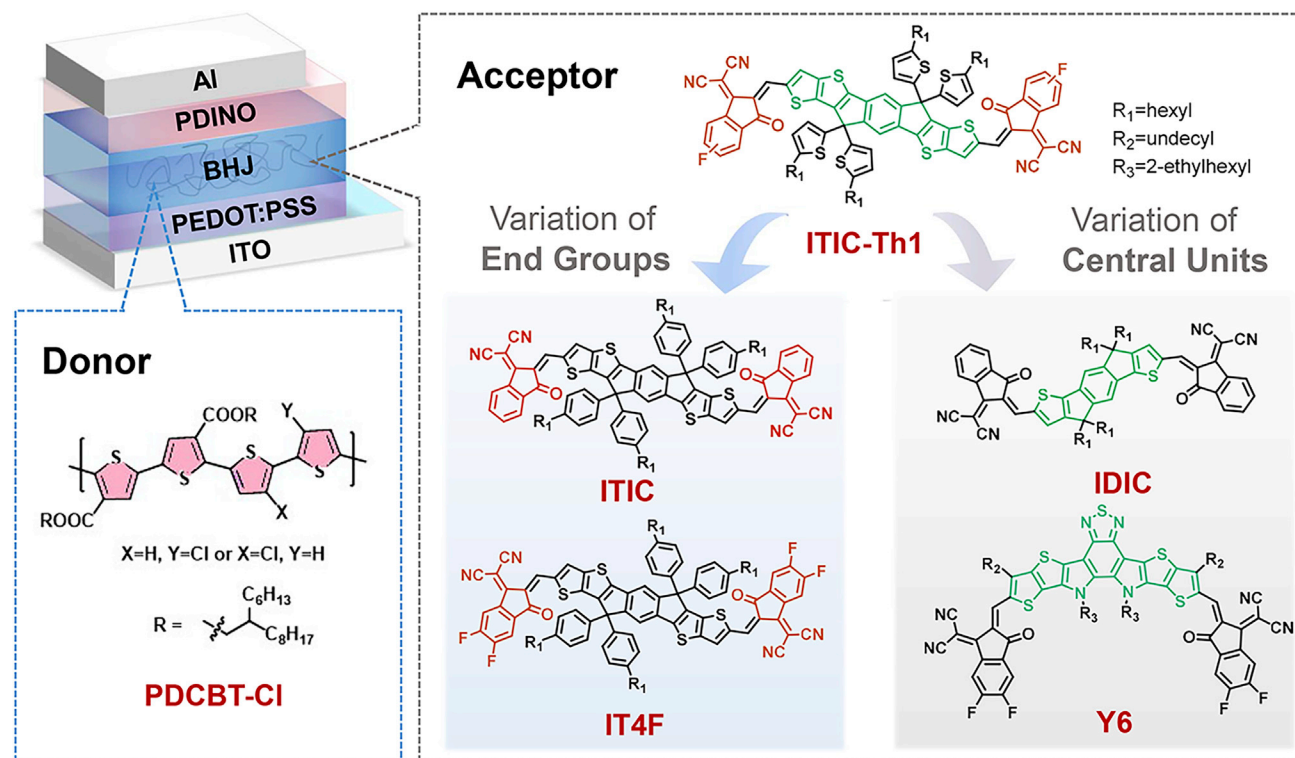


Figure 1. Device Architecture and Chemical Structures of the PT Derivative PDCBT-Cl and Five Representative Nonfullerene Acceptors with the Variation of End Groups (ITIC, ITIC-Th1, and IT4F) and Central Units (ITIC-Th1, IDIC, and Y6)

and Y6 as acceptor using conventional device architecture of indium oxide (ITO)/PEDOT:PSS/PDCBT-Cl:acceptor/PDINO/Al. The active layers based on different nonfullerene acceptors were processed by chloroform with a total concentration of 18 mg/mL and the optimal weight ratio of PT to acceptor is 1:1. Upon solvent vapor annealing with carbon disulfide for 2 min, all devices achieved the optimal performance. Their typical current density-voltage (*J*-*V*) curves are illustrated in Figure 2A and the corresponding device parameters are given in Table 1.

The best performing devices based on ITIC-Th1, ITIC, IT4F, IDIC, and Y6 showed V_{oc} values of 0.94, 1.02, 0.89, 0.95, and 0.88 V, respectively, which can be readily explained by the lowest unoccupied molecular orbital (LUMO) energy levels of these acceptors.^{5,22–25} The average J_{sc} values of these devices range from ~2 to ~18 mA cm⁻². Figure 2B, which exhibits the external quantum efficiency (EQE) as a function of the acceptor, reveals that the blend based on ITIC-Th1 produces the most efficient device and the integrated J_{sc} values are in accordance with the values from *J*-*V* data. Despite the broadest photoresponse-range, PDCBT-Cl:Y6 delivers a very disappointed J_{sc} among these blends. Conversely, ITIC-Th1-based devices give the highest J_{sc} value. The maximum FF values of these devices follows the order of ITIC-Th1 (70.0%) > ITIC (65.9%) > IDIC (65.8%) > IT4F (60.0%) > Y6 (27.1%). As a result, the champion photovoltaic performance (12.11%) is achieved for ITIC-Th1-based devices, while Y6-based devices show an extremely low PCE of 0.48%. The devices based on ITIC, IT4F, and IDIC exhibit moderate performance with PCEs of 7.25%, 9.05%, and 9.44%, respectively. The significant difference in solar cell performance of the five PDCBT-Cl:nonfullerene blends motivated us to understand the morphological origins from the viewpoints of mixing thermodynamics and film-

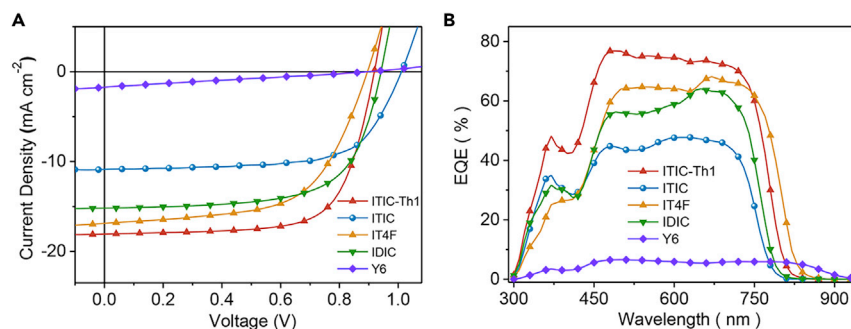


Figure 2. Photovoltaic Performance of PDCBT-Cl:Acceptor Systems

(A) Current density-voltage (J-V) characteristics of the OPVs under the optimized conditions.

(B) EQE curves of the OPVs under the optimized conditions.

See also [Tables S5](#) and [S6](#).

solidification kinetics, which are the pivotal factors that impact the film morphology and thus determine the device performance.

Crystallinity and Molecular Order Of PT:Nonfullerene Blends

The crystallinity and melting temperature of PDCBT-Cl blended with different acceptors were investigated by DSC. The first cooling curves with various acceptor amounts are illustrated in [Figure S1](#) and the corresponding enthalpies of fusion are shown in [Figure S2](#). Neat PDCBT-Cl exhibited a moderate enthalpy of ~ 16 J/g. As the acceptor amount increases, the crystallization enthalpy (directly correlated to the total energy that is required to break the intermolecular forces) decreases, indicating that the acceptors interact with PDCBT-Cl and destroy the crystallites of PDCBT-Cl.²⁶ A noticeable decline in the crystallization enthalpies was observed for PDCBT-Cl:Y6 and PDCBT-Cl:IT4F-based blends, while a relatively small impact of ITIC-Th1 and IDIC-based blends was found for the crystallites of PDCBT-Cl upon blending.

In order to understand how the variation of acceptor molecules leads to such a large difference in photovoltaic performance among these OPV devices, we examined molecular packing and crystalline texture in the blend films by using GIWAXS²⁷ at the Advanced Light Source. The GIWAXS patterns and the 1D profiles were shown in [Figures 3A–3F](#). In addition, the peak-fitting results and corresponding data were displayed in [Figure S3](#); [Table S1](#). The neat PDCBT-Cl film (see [Figure S4](#)) exhibits pronounced (010) diffraction peak at approximately $1.73 \text{ \AA}^{-1}$ in both out-of-plane (q_z) and in-plane (q_{xy}) directions with π - π stacking distance of $\sim 3.6 \text{ \AA}$, indicating that edge-on and face-on orientations coexist in the film. For the neat acceptor films (see [Figure S5](#)), the (010) diffraction peaks located at $1.77\text{--}1.81 \text{ \AA}^{-1}$ are along the q_z direction, which are characteristics of face-on π - π stacking. Among them, IDIC and ITIC-Th1 exhibit a pronounced (010) peak at $q_z = 1.8 \text{ \AA}^{-1}$, while the ITIC shows a much weaker peak and IT-4F was nearly amorphous.²⁸ For all of these blend films, a preferential face-on molecular orientation is observed as the (010) diffraction peaks are more intense in the q_z direction than those in the q_{xy} direction. The π - π coherence lengths (L_c) were quantified by the full width at half-maximum of the (010) diffraction peak along q_z direction via Scherrer Equation, and paracrystallinity parameters (g) can be also estimated.²⁹ The L_c values were 18.9, 21.8, and 26.9 $\text{\AA}$ for ITIC, IT4F, and Y6-based blends, and their respective g parameters were 11.5%, 10.8%, 9.7%. Compared to PDCBT-Cl:ITIC, PDCBT-Cl:ITIC-4F, and PDCBT-Cl:Y6 blend films, PDCBT-Cl:ITIC-Th1 and PDCBT-Cl:IDIC blend films exhibited more ordered π - π stacking with much higher L_c values of 31.4 and 43.5 $\text{\AA}$, along with the smaller g parameters of 9.0% and 7.7%, respectively. Distinct (100) diffraction peaks

Table 1. Photovoltaic Performance of the OPVs Based on PDCBT-Cl:Acceptor Blends

Acceptor	V _{oc} (V)	J _{sc} (mA/cm ²)	FF (%)	PCE (%) ^a
ITIC-Th1	0.93 ± 0.01 (0.94)	18.1 ± 0.2 (18.3)	68.5 ± 1.4 (70.0)	11.95 ± 0.13 (12.11)
ITIC	1.01 ± 0.01 (1.02)	10.6 ± 0.1 (10.8)	64.8 ± 1.0 (65.9)	7.05 ± 0.17 (7.25)
IT4F	0.88 ± 0.01 (0.89)	16.9 ± 0.3 (17.2)	58.0 ± 1.5 (60.0)	8.73 ± 0.29 (9.05)
IDIC	0.94 ± 0.01 (0.95)	15.0 ± 0.1 (15.2)	64.2 ± 0.9 (65.8)	9.12 ± 0.13 (9.44)
Y6	0.87 ± 0.01 (0.88)	2.0 ± 0.1 (2.2)	26.8 ± 0.2 (27.1)	0.46 ± 0.01 (0.48)

^aStatistical results are obtained from over 15 devices. The maximum results are listed in the parentheses.

coming from the lamellar packing of PDCBT-Cl are observed in both q_z and q_{xy} directions at $\sim 0.29 \text{ \AA}^{-1}$ for the blend films except PDCBT-Cl:Y6 system, in which no obvious polymer lamellar packing exists. In addition to good lamellar packing of PDCBT-Cl, blend films of PDCBT-Cl:ITIC-Th1 and PDCBT-Cl:IDIC also show pronounced (100) peaks at $\sim 0.29 \text{ \AA}^{-1}$ and the high orders of (100) diffraction peaks along q_z and q_{xy} arising from the acceptor molecules. Therefore, both donor polymer and acceptor molecules in PDCBT-Cl:ITIC-Th1 and PDCBT-Cl:IDIC blend films displayed relatively highly ordered molecular packing, which could be beneficial for charge transport. In particular, as evidenced by the multiple diffraction features in the q range of $0.3\text{--}0.7 \text{ \AA}^{-1}$ and highest (010) diffraction intensity combined with the smallest g parameter, PDCBT-Cl:IDIC is the most ordered system among the five blends. By contrast, the PDCBT-Cl:Y6 film shows relatively weak (100) diffraction at $\sim 0.37 \text{ \AA}^{-1}$ from lamella packing of Y6,⁵ and there is no obvious (100) diffraction from the acceptors existed in PDCBT-Cl:ITIC and PDCBT-Cl:IT4F films, implying poor lamellar packing of the acceptor molecules. The results (Figure S6) obtained at the Shanghai Radiation Synchrotron Facility (SSRF) demonstrated similar trends. Although these observations could partially explain the difference in the photovoltaic performance of these PDCBT-Cl:acceptor blend systems, other morphological parameters, such as length scales of the compositional domains (long periods) and relative domain purity in the blend films should also contribute to the variations in the device efficiency, which will be discussed below.

Mesoscale Domain Characteristics Of PT:Nonfullerene Blends

RSOXS³⁰ offers a quantitative understanding of the mesoscale morphology characteristics of these PDCBT-Cl:acceptor blend films under the optimal processing conditions. As represented in Figure 4A, the 1D RSOXS profiles of all films were dominated by a single lognormal peak.³¹ Based on the profile fitting with well-defined lognormal functions, the long period (L) of the phase separation (center-to-center domain spacing) can be calculated from the q location of the predominant peak using the equation: $L = 2\pi/q$. The L values are 68.3, 75.1, 51.5, 142.8, and 64.8 nm for PDCBT-Cl:ITIC-Th1, PDCBT-Cl:ITIC, PDCBT-Cl:IT-4F, PDCBT-Cl:IDIC, and PDCBT-Cl:Y6 blend films, respectively (Figure 4B). The PDCBT-Cl:IDIC film exhibited oversized phase separation with long period ~ 2 times larger than that of the PDCBT-Cl:ITIC-Th1 film. Obviously, the long periods of all blend films are larger than the typical exciton diffusion length (~ 20 nm). Consequently, suitably mixed domains are needed to provide enough contacts (distributed “interfaces”) between the donor and the acceptor molecules for exciton dissociation.^{32,33}

Herein, the comparison of composition variations, which are monotonically but not linearly related to the relative domain purity, was made by assessing the square root of integrated scattering intensity over the entire q range of the measurement,³⁴ a measure that is proportional to the standard deviation σ of the composition provided that orientation contrast is negligible. The mathematical expression of σ is

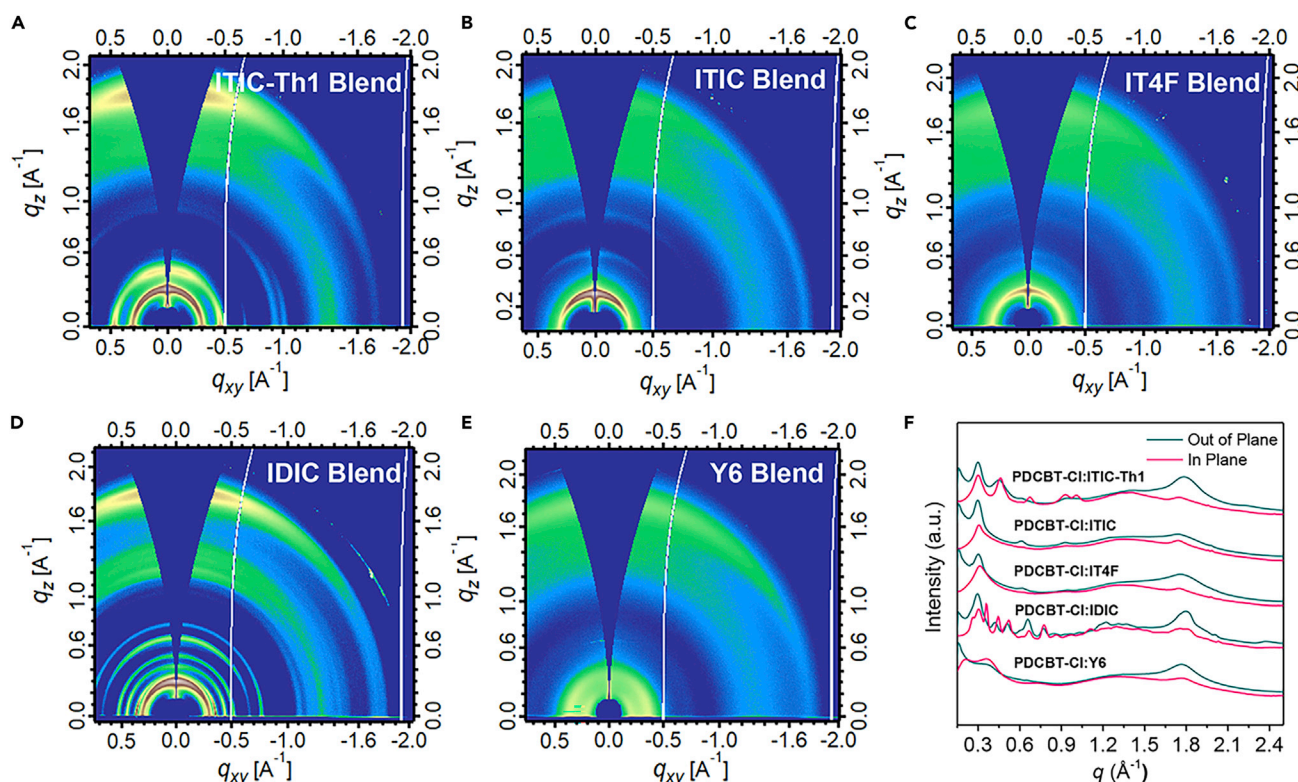


Figure 3. GIWAXS 2D Patterns and 1D Profiles

(A–F) 2D GIWAXS patterns for (A) PDCBT-Cl:ITIC-Th1; (B) PDCBT-Cl:ITIC; (C) PDCBT-Cl:IT4F; (D) PDCBT-Cl:IDIC; (E) PDCBT-Cl:Y6 blend films under the optimal conditions; and (F) 1D out-of-plane and in-plane profiles derived from (A)–(E). See also [Figures S3–S6](#)

briefly described in [Figure S7](#); [Equation S1](#). As illustrated in [Figure 4B](#), the relative σ (“domain purity”) of these blend films follows the order of PDCBT-Cl:ITIC (1) > PDCBT-Cl:IDIC (0.87) > PDCBT-Cl:ITIC-Th1 (0.79) > PDCBT-Cl:IT-4F (0.63) > PDCBT-Cl:Y6 (0.38). It is obvious that the PDCBT-Cl:ITIC and PDCBT-Cl:Y6 blend films exhibit the highest and lowest domain purity, respectively. Another interesting observation is that the best performing PT:nonfullerene system (i.e., PDCBT-Cl:ITIC-Th1) only exhibited a moderate domain purity. The apparent difference of the actual morphology between PDCBT-Cl:ITIC-Th1 and PDCBT-Cl:Y6 systems can also be observed from the real-space results of atomic force microscopy (AFM) measurements ([Figure S8](#)). The results from the real-space morphology characterizations further support our previous RSoXS analysis. The implications are further discussed below.

Thermodynamic Mixing Of PT:Nonfullerene Blends

To reveal the fundamental factors dominating the domain characteristics, thermodynamic molecular interactions between PDCBT-Cl and the acceptor materials were investigated. The Flory-Huggins interaction parameter (χ_{aa}) is known as a fundamental metric of degree of mixing.³⁵ The lower χ_{aa} value indicates better miscibility between two components, which often leads to lower domain purity. We employed DSC to monitor the melting point depression of the blends with various PDCBT-Cl:acceptor ratios.³⁶ As shown in [Figures 5](#) and [S10](#), the melting temperatures of PDCBT-Cl blending with different acceptors are gradually reduced as the acceptor amount increases. More specifically, we can obtain χ_{aa} at the melting temperature

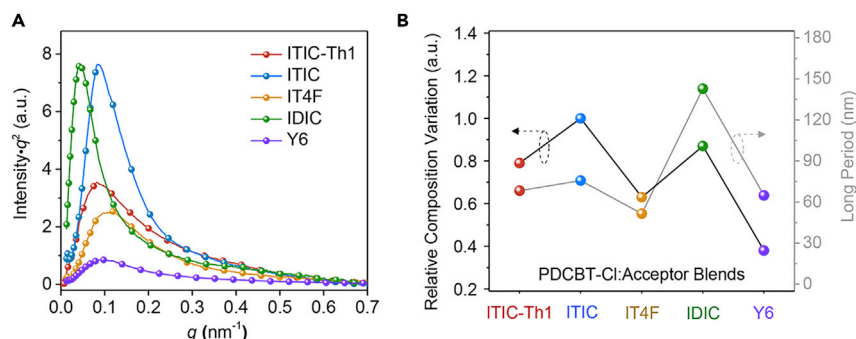


Figure 4. Thickness and Absorption Normalized RSoXS Profiles Acquired at 283.2 eV

(A) 1D profiles of the blends processing with optimized annealing time.

(B) Plots of long periods and relative standard deviation of the composition (σ) extracted from RSoXS profiles of the blends in the optimized devices.

(T_m) by fitting the depression of melting point and acceptor's volume fraction to a quadratic function using Nishi-Wang framework and Equation 1.³⁷

$$\frac{1}{T_m} - \frac{1}{T_m^0} = \frac{R}{\Delta H_f} \frac{v_m}{v_s} (\phi_s - \chi_{aa} \phi_s^2) \quad (\text{Equation 1})$$

where T_m is the melting point of the crystalline polymer in a blend, T_m^0 and ΔH_f refer to the melting temperature and melting enthalpy of pure crystalline polymer, v_m is the monomer molar volume of polymer and v_s is the molar volume of the acceptor, ϕ_s represents the volume fraction of the acceptor, and R is the ideal gas constant. Fitting the melting point data to Equation 1 yields $\chi_{aa} = 0.68, 1.73, 0.34, 1.75, 0.19$ for PDCBT-Cl blending with ITIC-Th1, ITIC, IT4F, IDIC, and Y6, respectively. We also study the polymer:acceptor miscibility at room temperature (RT) through surface tensions (γ), calculated by the contact angles of the neat films of PDCBT-Cl and the acceptors with two distinct solvents.^{34,38,39} According to an empirical relation⁴⁰ $\chi_{aa} \propto (\sqrt{\gamma_{donor}} - \sqrt{\gamma_{acceptor}})^2$, the relative $\chi_{aa}(RT)$ values of these blend systems follow the order of PDCBT-Cl:IDIC > PDCBT-Cl:ITIC > PDCBT-Cl:ITIC-Th1 > PDCBT-Cl:IT4F > PDCBT-Cl:Y6, which agrees well with the trend of $\chi_{aa}(T_m)$ values obtained from DSC measurement (Table S2; Figure S11).

It has been previously shown that the acceptor concentration in the polymer-rich mixed domains has to be frequently at or above the electron percolation threshold in order to avoid geminate recombination and facilitate charge extraction.^{28,41–43} The ability to reach such a concentration depends on the miscibility and casting kinetics. Regarding miscibility, it can be seen that with the increasing number of F atoms in end groups, the nonfullerene acceptors exhibit enhanced miscibility with PDCBT-Cl. The χ_{aa} values of PDCBT-Cl:ITIC and PDCBT-Cl:IDIC blend systems were much higher than those of other blend systems. The low miscibility should be the driving force for the formation of the relatively high purity of mixed domains in PDCBT-Cl:ITIC and PDCBT-Cl:IDIC blend films. In addition, the severe phase separation in PDCBT-Cl:IDIC blend films should be also related to the hypo-miscibility between the two components. In contrast, the high miscibility between PDCBT-Cl and Y6 induced impure domains and poor molecular packing in the blend film. The results from the thermodynamic analysis agree quite well with the AFM observations of the as-cast films (Figure S9). It was noted that the order of χ_{aa} values is not completely identical to that of the domain purity of mixed domains in these blend films under the optimal conditions, indicating that kinetic factors also play a

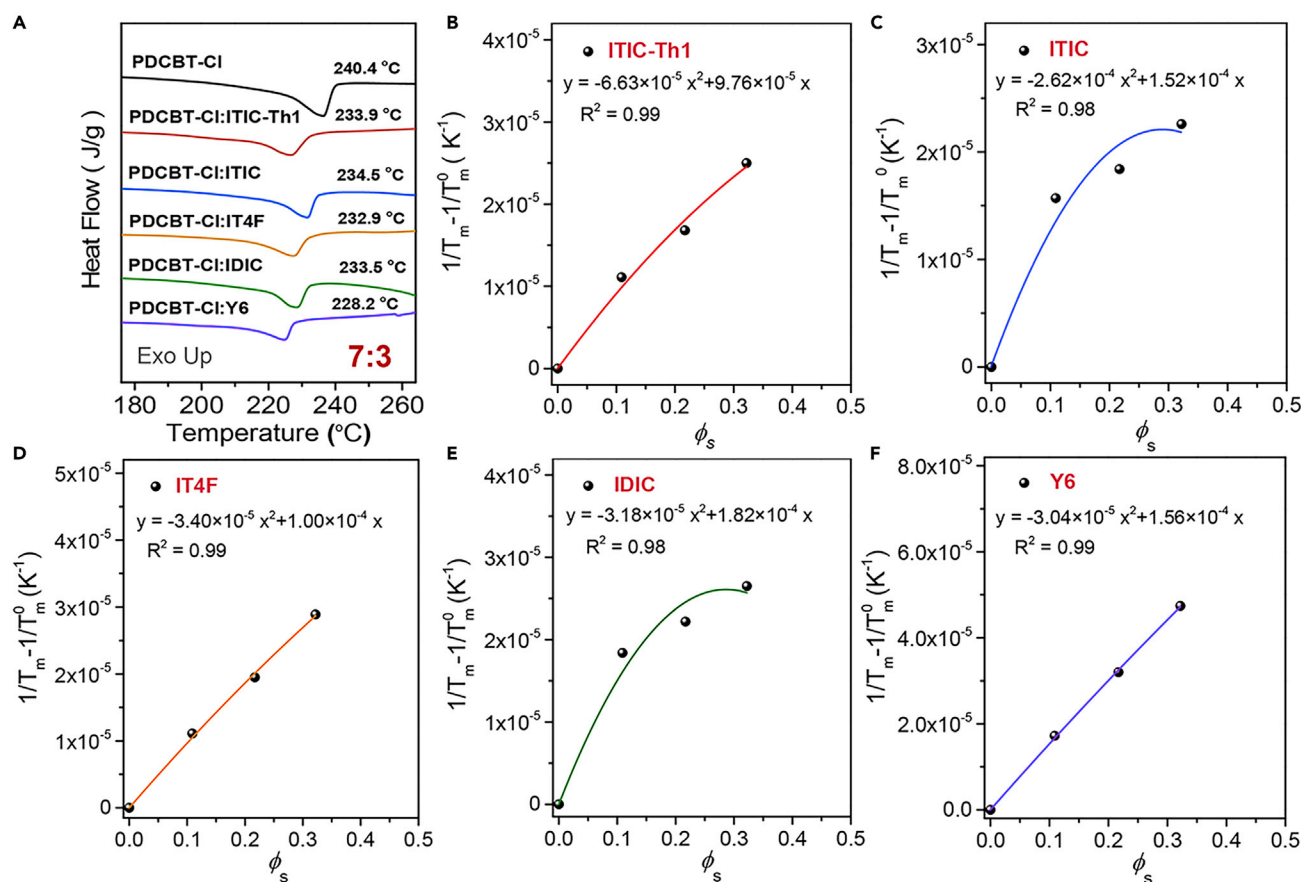


Figure 5. Estimation of the Miscibility According to the Melting Point Depression

(A) Melting point depression of PDCBT-Cl blending with various acceptors.

(B–F) Estimates of the donor:acceptor miscibility from measurements of the melting point depression as a function of acceptor's volume fraction, ϕ_s , obtained from DSC experiments fitting the Nishi-Wang formalism (Equation 1): (B) ITIC-Th1, (C) ITIC, (D) IT4F, (E) IDIC, (F) Y6. See also Figure S10.

role in determining the device morphology. For example, the PDCBT-Cl:IDIC blend displays higher χ_{aa} value than PDCBT-Cl:ITIC, while lower domain purity is observed for the IDIC blend. As previously demonstrated,^{40,44} the presence of solvent or the solubility difference usually affects the kinetic process, for instance, liquid-solid and liquid-liquid phase separation, and in turn manipulates the composition of mixed phase. Therefore, the spin-cast blend films are often kinetically quenched and likely in metastable state rather than thermodynamic equilibrium state even after post treatments,²⁸ and the content of crystalline components in blends also have an effect on miscibility and domain purity.⁴¹ These aspects are likely to play a role here too and will be discussed below.

According to the results of GIWAXS and DSC analyzed above, PDCBT-Cl crystallites exist in blend films for all systems except Y6-based blends. Thus, solid (crystalline component)-liquid phase transition occurs in addition to amorphous-amorphous phase segregation.⁴⁵ Based on a self-consistent model of crystalline-amorphous blends developed by Kyu et al.,^{34,41,46} crystalline-amorphous interaction parameter (χ_{ca}) acts as a supplement to χ_{aa} in describing crystalline-amorphous systems and also plays a key role in determining the morphology formation. Following Kyu, we endeavor to estimate χ_{ca} using an equation: $\chi_{ca} = S \frac{\Delta H}{RT_m}$, where S is the slope of

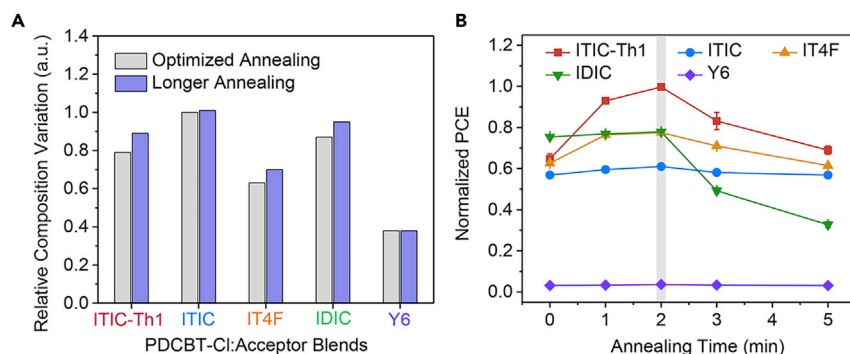


Figure 6. RSoXS Purities and Morphology-Performance Relations

(A) Plots of relative composition variations of PDCBT-Cl:acceptor blends with optimized annealing (2 min) and longer annealing (1 h). See also Figure S13.

(B) Plots of normalized PCEs for PDCBT-Cl:acceptor blends as a function of solvent vapor annealing time. The gray band indicates the optimum annealing time for PDCBT-Cl:acceptor blends. Error bars are obtained from over 10 devices. See also Figure S16.

$\left(1 - \frac{T_m}{T_m^0}\right)$ against $(1-\phi)$ plot. Herein, the S values of PDCBT-Cl:acceptor systems are obtained by linear fitting the depression of melting point and acceptor's volume fractions, as shown in Figure S12. For ITIC and IDIC-based systems, the linearity displays a sharp downward departure even under a high volume fraction of polymer. The reduction of the linear portion is mainly attributed to the decreased donor:acceptor miscibility, which will increase the interference from liquid-liquid phase separation.⁴⁶ Hence, the poor linear correlation prevents the further determination of χ_{ca} values for these two systems. In contrast, ITIC-Th1- and IT4F-based blends exhibit relatively significant linear dependence. The values of S for PDCBT-Cl:ITIC-Th1 and PDCBT-Cl:IT4F blend systems are 0.058 and 0.055, respectively. Therefore utilizing the values extracted from DSC of pure PDCBT-Cl samples, χ_{ca} can be calculated to be 0.18 for PDCBT-Cl:ITIC-Th1 and 0.17 for PDCBT-Cl:IT4F. The positive values of χ_{ca} indicate a repulsive interaction between PDCBT-Cl crystallites and amorphous domains, which facilitate the formation of pure crystalline phases. The slightly increased repulsive interaction parameter for PDCBT-Cl:ITIC-Th1 system will induce the formation of a larger fraction of PDCBT-Cl crystallites, which can also be speculated from the GIWAXS results that the volume normalized scattering intensity of (100) peaks coming from the lamellar packing of PDCBT-Cl in the ITIC-Th1-based blend is ~ 1.4 times higher than that of IT4F system.

Kinetic Quenching Of PT:Nonfullerene Blends

To thoroughly understand the actual morphology in devices, we further evaluated the extent of kinetic quenching of these films. The blend films subject to extensive solvent vapor annealing were used to investigate the composition of the mixed domains as they traverse the percolation threshold⁴⁷ and evolve toward the equilibrium state. As shown in Figures 6A and S13, annealed PDCBT-Cl:ITIC-Th1, PDCBT-Cl:IT4F, and PDCBT-Cl:IDIC blend films exhibit coarsened domains and obviously increased domain purity in comparison to the films in the optimized OPVs (detailed data summarized in Tables S3 and S4), suggesting that even the composition of the mixed domains in the optimized blend films have been quenched away from local binodal equilibrium. The domain composition of high χ_{aa} system PDCBT-Cl:ITIC was almost unchanged, which implied that the system approached the local thermodynamic equilibrium during processing. For the PDCBT-Cl:Y6 blend system, the

domain purity was almost the same after solvent vapor annealing for 1 h. Accordingly, the hyper-miscibility system PDCBT-Cl:Y6 remained in a poorly phase-separated state in the optimized OPVs, with the mixed amorphous domains possibly in the one-phase state and there was no scattering intensity observed arising from PDCBT-Cl crystals.

To further analyze the degree of kinetic quenching for different blends, the composition dependence of electron mobility data (Figures S14 and S15) was first tested. For all five blends, there is a sharp increase in electron mobility when the acceptor composition in the blends is in the range of 20%~25% by volume, which should be the percolation threshold (ϕ_p) for electron transport in the blends.⁴⁸ Subsequently, we acquired the device performance as a function of solvent vapor annealing time. As presented in Figures 6B and S16, after annealing for 2 min, all devices based on different acceptors obtained the optimal performance. Among them, ITIC-Th1-, IDIC-, and IT4F-based devices showed declines of FF, J_{sc} , and PCE beyond their optimum performance, which correspond to the percolation conditions. Compared with ITIC-Th1 and IDIC devices, IT4F devices exhibited a weaker dependence on annealing time. Interestingly, the performance of IDIC-based blend showed a slight variation in the first 2 min, which may be related to the crystallinity and diffusion properties of IDIC. Regardless of the reason, this phenomenon implies that the as-cast IDIC film is relatively close to the percolation condition, whereas the ITIC-Th1 is far from the percolation condition. Once the annealing time is more than 2 min, both ITIC-Th1 and IDIC systems start to over-purify. Strong time dependence of the photovoltaic parameters can be observed in the PDCBT-Cl:ITIC-Th1 blend, which is the conclusive evidence that the mixed domains are indeed quenched in the optimized devices. The performance of PDCBT-Cl:ITIC-Th1 and PDCBT-Cl:IT4F initially improved as the morphology approached percolation conditions; however, their performance rapidly degraded after they traversed the percolation condition, which is due to the formation of over-pure domains. Unlike the above behaviors, solvent vapor annealing time shows little effect on the performance of the PDCBT-Cl:ITIC blend, indicating the film morphology's most likely approach to the equilibrium mixed composition even in as-cast devices. Ye et al.²⁸ reported that there exists a strong molecular weight dependence of the quench degree for PM6:IT4F systems with hypo-miscibility. The hypo-miscible PDCBT-Cl:ITIC-Th1 may have similar effects, while the hyper-miscible systems, such as PDCBT-Cl:IT4F and PDCBT-Cl:Y6 likely have a weak dependence on molecular weight. Using the same batch of PDCBT-Cl with relatively high molecular weight for device fabrication can exclude the impact of molecular weight. Based upon the results on film morphology under the optimal conditions in combination with the photovoltaic performance under varying annealing time, the extents of kinetic quenching for these blend systems and further purification upon vapor annealing can be exemplified and delineated in the phase diagram (Figure 7A). Despite the presence of crystalline phase in blend films, the χ_{aa} of each system in the amorphous phase diagram will not be affected by solvent vapor annealing. The extent of kinetic quenching will likely vary with time during the solvent vapor annealing process. Concerning this point, we only qualitatively determine the relative positions in Figure 7A.

Morphology-Function Relationship of PT:Nonfullerene Blends

The above characterizations collectively allow us to connect device performance with morphological properties. For the PDCBT-Cl:IT4F blend, the amorphous nature of IT4F results in difficulties to regulate its morphology with desired phase separation, although IT4F displays moderate miscibility with PDCBT-Cl. In contrast, driven by hypo-miscibility of PDCBT-Cl and ITIC, over-purified domains can form easily

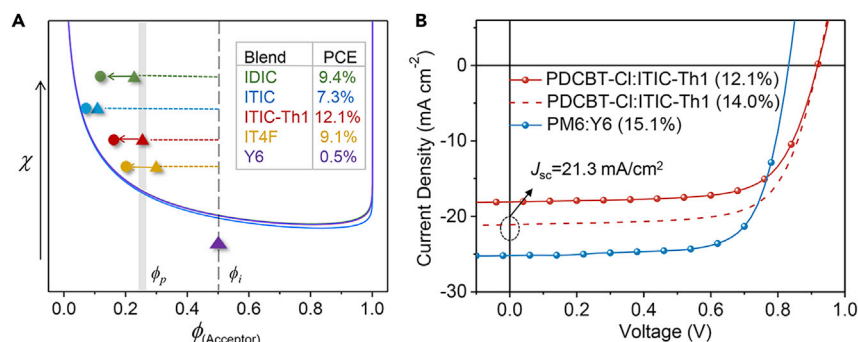


Figure 7. Phase Diagram Of PDCBT-Cl:Acceptor Systems and J-V Curves Of PDCBT-Cl:ITIC-Th1 System

(A) Schematic illustration of the amorphous phase diagram based on PDCBT-Cl:acceptor blends in relation to the electron transport percolation threshold. The gray shadow marks the measured percolation threshold (ϕ_p) for electron transport (see Figure S15). Triangles and dots represent the states in optimized devices and long-time annealed devices, respectively. The relative heights of the various triangles or dots are displayed according to the trend of the χ_{aa} results from melting temperature depression method and contact angle method. The arrows represent the evolution of the composition of mixed phase during further solvent annealing. ϕ_i indicates the acceptor volume fraction of donor:acceptor blends in the cast solutions. The χ - ϕ phase diagrams of the PDCBT-Cl:acceptor blends were constructed according to actual molecular weights and densities. The acceptor molecules with low molecular weight tend to form almost pure domains and the amorphous mixed domains are dominated by the polymer component.

(B) J-V curves of the optimized PDCBT-Cl:ITIC-Th1 devices. PM6:Y6 devices were reproduced in our lab and were used for comparison. The dashed line shows the calculated J-V curve of PDCBT-Cl:ITIC-Th1 devices if the maximum EQE can reach the same value of PM6:Y6 devices reported by Zou et al.⁵

leading to the composition of mixed phase away from the percolation threshold even for the fresh film. The PDCBT-Cl:ITIC blend film with the highest domain purity lacks percolation pathways for electron transport,⁴¹ resulting in the low J_{sc} ($\sim 10 \text{ mA cm}^{-2}$). Similarly, the oversized phase separation in the blend of PDCBT-Cl:IDIC is unfavorable for exciton diffusion and results in severe exciton recombination, which should be responsible for the relatively low J_{sc} in the OPVs. Due to the proper miscibility between PDCBT-Cl and ITIC-Th1, the composition of its mixed phase can be quenched close to the percolation threshold through precise post-treatment regulation. The optimally quenched morphology contributed to the highest J_{sc} and FF in the PDCBT-Cl:ITIC-Th1-based OPVs. As a result, PDCBT-Cl:ITIC-Th1 delivered a considerable PCE over 12%, which is among the best values of PT:nonfullerene solar cells reported so far. In stark contrast, the Y6-based blend remained in the one-phase state as showcased in Figure 7A due to the high miscibility between Y6 and PDCBT-Cl, resulting in a deficiency of phase separation. The resulting low domain purity and poor crystallinity could induce severe geminate and bimolecular recombination aided by poor mobility and charge extraction, which should be the key reason for the inferior photovoltaic performance ($\sim 0.5\%$) of PDCBT-Cl:Y6. Besides the influence of miscibility, the substrate also plays a crucial role in determining the morphology due to the effect on the thermodynamic behavior of systems.^{49–52} The impact of substrate especially for thicker films is mainly reflected in the vertical phase separation structures. However, the specific analysis of this aspect is beyond the scope of this study.

Beyond being an influence on device performance, donor:acceptor miscibility and the extent of quenching also impact the morphological stability and performance of aged devices.^{41,47} As a consequence, the shelf-aging stability of unencapsulated

devices based on five blends was tested in an argon-filled glove box at RT. As shown in Figure S17, the IT4F- and IDIC-based devices exhibit relatively poor stability with ~20% degradation after 100 h of aging in an inert atmosphere, primarily due to the fact that morphology has been quenched far from the equilibrium state. ITIC-based devices display better stability with degradation less than 10% mainly because the morphology is close to the binodal composition, which represents the thermodynamic equilibrium. ITIC, however, never yields a favorable morphology close to the percolation threshold. In contrast, for the Y6-based system located in the homogeneous region, the device performance has barely changed after 100 h of aging (Figure S18). This experimental observation also confirms recent miscibility-function and stability models taking into account the need to have percolation when the domain size is larger than the exciton diffusion length.^{28,34,41,47,53}

Having fully understood the miscibility and crystallinity of the PDCBT-Cl:acceptor blend films by DSC, GIWAXS, RSoXS, and contact angle characterizations, it is possible now to illustrate the schematic morphology (Figure S19). As demonstrated in the PDCBT-Cl:ITIC-Th1 case, moderate miscibility between PTs and acceptors are conducive to adjust the morphology after post-treatment, resulting in high performance. Conversely, hyper-miscibility (low χ_{aa}) systems, such as PDCBT-Cl:Y6 need to be avoided, as it will result in insufficient phase separation that usually leads to increased charge recombination. Conversely, hypo-miscible blends like PDCBT-Cl:IDIC often have enough driving force for the formation of oversized and excessively pure domains, resulting in difficulties to properly quench the morphology to the percolation threshold.²⁸ Hypo-miscible systems might also lack device stability. Beyond driving forces, if the χ parameter rankings are solid, the case of the ITIC system would further indicate that purely kinetic factors are also important, as the ITIC-based system could not be quenched to the percolation threshold despite its higher miscibility compared with IDIC-based devices and the use of a quickly evaporating casting solvent. Based on integrated considerations of device performance and stability, the best quenched composition needs to be kept close to the percolation threshold, as illustrated in Figure 7A.

Design Rules toward More Efficient PT:Nonfullerene Solar Cells

Based upon the above results, neither nonfluorinated nor fully fluorinated acceptors are conducive to form favorable mixing that provides high performance and stable morphologies as well as high and matched crystallinity with PDCBT-Cl. To further improve the performance of PT-based OPV devices, novel design rules for matching PT and nonfullerene acceptors need to be established. As the degree of mixing and crystallinity in the optimized blend films are heavily related to the molecular interactions, χ_{aa} and χ_{ca} parameters can assist with the identification of high performance and low-performance systems. By simply determining these physical parameters of material systems, we can, therefore, exclude low-performance pairs and importantly screen promising PT:nonfullerene pairs. From the perspective of molecular structure, the central units largely determine the shape of molecules and have a great effect on crystallinity.⁵⁴ Steric hindrance of non-conjugated side chains (ITIC, IT4F) and meniscus-like molecular skeleton (Y6) are not conducive for the formation of intermolecular compact stacking. Conversely, the molecules with planar and linear structure shapes often have higher crystallinity like IDIC.^{9,55} In addition, the introduction of F atoms to the end groups of acceptors is an effective method to promote their miscibility with PDCBT-Cl, as we demonstrated above, and will not cause an apparent sacrifice of crystallinity.^{56,57} Thereby introducing mono-fluorinated end groups to IDIC might be a valid concept to increase the miscibility with PDCBT-Cl.

Combined with high crystallinity, the fluorinated IDIC will be more miscible with PDCBT-Cl and might produce a more efficient performance in the OPV device.

Lastly, Y6, as a rising-star nonfullerene acceptor in the field, delivers respectable efficiencies over 16% by pairing with many donor polymers.^{2,4,5,58–61} However, the performance of PDCBT-Cl:Y6 is ~30 times lower compared with that of recent record-efficiency PM6:Y6. Lowering the miscibility by engineering the side chains of PDCBT-Cl might be a feasible approach to improve the performance of PT:Y6-based devices according to a systematic study of side-chain effect by Huang and co-workers.⁶² Alternatively, introducing cyano groups to the thiophene units of PT might also work, as a most recent study⁶³ revealed that the attachment of cyano groups to polymer backbones can lower the donor:acceptor miscibility and improve the performance by a factor of ~5. Besides, the performance of highly miscible systems, such as PDCBT-Cl:Y6 can be improved by designing new device configurations. For instance, it was reported that sequential deposition can form quasi-bulk heterojunction with appropriate length scales⁶⁴ and better crystallinity, which outperforms its control device, based on conventional bulk heterojunction. Furthermore, as Y6 exhibits a hyper-miscibility with PDCBT-Cl, Y6 can be utilized as a third component for realizing highly efficient and stable ternary devices in cases where the host binary blend is hypo-miscible and has a tendency to over-purify.⁶⁵ It is also fairer to compare our best performance system PDCBT-Cl:ITIC-Th1 with the recent record-efficiency system PM6:Y6 from a morphology perspective rather than based purely on PCE values, as the theoretically maximum achievable J_{sc} values of PDCBT-Cl:ITIC-Th1 and PM6:Y6⁵ are very different, due to their different bandgaps. As showcased in Figure 7B, PDCBT-Cl:ITIC-Th1 will probably deliver a high PCE of ~14% if the morphology of PDCBT-Cl:ITIC-Th1 is properly tuned with maximum EQE comparable to that of PM6:Y6.

Conclusions

In summary, aiming at establishing a guideline for materials matching for PT:nonfullerene systems, we have thoroughly characterized the miscibility, molecular ordering, and film morphology of binary blends based on a representative PT derivative PDCBT-Cl and five nonfullerene acceptors with varied end groups or central units, named ITIC-Th1, ITIC, IT4F, IDIC, Y6. The relationship between chemical structures of nonfullerene acceptor and device performance for PT:nonfullerene system was revealed from the viewpoints of mixing thermodynamics and film-solidification kinetics. Relatively high crystallinity of ITIC-Th1 and IDIC facilitated the charge transport in the blend films along with the PDCBT-Cl crystallites. On the contrary, hypo-miscibility between PDCBT-Cl and IDIC provided the driving force for the formation of large phase separation, which caused a lower FF and J_{sc} . ITIC, IT4F, and Y6 exhibited a similarly poor crystallinity but a remarkably different miscibility with PDCBT-Cl. In particular, Y6 was most miscible with PDCBT-Cl, indicating that the blend largely remains in one-phase state, and hence the device possessed an extremely low J_{sc} , and FF. Notably, the moderate miscibility between PDCBT-Cl and ITIC-Th1 with the mono-fluorinated end groups helped the formation of suitable domain purity. The final acceptor concentration in the mixed phase of the PDCBT-Cl:ITIC-Th1 blend film can be locked in with an optimal composition close to the electron percolation threshold after proper post-treatment, thereby leading to the highest efficiency of over 12%. Overall, high molecular ordering for both donor and acceptor is needed in order to form compact and ordered stacking in films, and moderate mixing is required in the amorphous mixed phases. These features are required to realize high efficiency in PT:nonfullerene OPVs. In contrast, hyper-

miscibility between donor and acceptor makes it impossible for morphology to approach these desired features.

The spectacular failure of Y6 with PDCBT-Cl indicates that the field of OPVs has optimized only one branch of donor:nonfullerene acceptor systems, a process during which acceptors were optimized in an iterative trial-and-error process to work well with the favorite donors at the time, namely FTAZ and its derivatives, PTB7 and its derivatives, and in particular PM6 and its derivatives, whose developments themselves were really driven by the optimization of performance with fullerenes. We note that all these classic donors are based on benzodithiophene (BDT) electron-donating moieties flanked by one or two thiophenes. From a thermodynamic perspective, the successful novel nonfullerene acceptor had to mimic or emulate the miscibility properties of fullerene. This view is supported by our prior observations that perylene diimide-based acceptors tend to be hyper-miscible in donors optimized for fullerenes.^{34,66} Successful optimization of different classes of donor polymers desirable for their costs or processing (water or alcohol) characteristics might not start or branch off from this existing main optimization branch, but might require a broader search for optimized structures, possibly with the aid of machine learning. A recent non-OPV example of such a search is the discovery of a novel class of antibiotics for super-resistant bacteria.⁶⁷ In short, classes of donors and acceptors might have simply been discarded based on their thermodynamic properties and resulting poor morphology, not based on their optical, energy level or transport properties. This realization provides optimism that cheaper materials, such as PTs and materials processed from water and alcohol mixtures might be able to close the performance gap to the current branch of optimized material pairs that exclusively rely on BDT-based donors paired with extended core, planar acceptors.

EXPERIMENTAL PROCEDURES

Resource Availability

Lead Contact

Further information and requests for resources and materials should be directed to and will be fulfilled by the Lead Contact, Long Ye (yelong@tju.edu.cn).

Materials Availability

This study did not generate new unique materials.

Data and Code Availability

The raw data and code supporting the current study will be made available from the Lead Contact on request.

Materials

The model polymer PDCBT-Cl (number-averaged molecular weight = 25.1 kDa, polydispersity index = 2.21) used in this work were synthesized according to our previous report.¹⁵ All acceptors (ITIC-Th1, ITIC, IT4F, IDIC, and Y6) and PDINO were purchased from Derthon Optoelectronic Materials Science Technology Co. Ltd. Anhydrous chloroform was supplied by Sigma-Aldrich and used without further purification. The full names of materials mentioned in this study are listed in [Supplemental Information](#).

DSC Measurements

DSC samples were dissolved in anhydrous chloroform and stirred at 45°C for 3 h. Then the solution was drop-cast wise on a glass substrate and kept in high vacuum for 2 h to remove the residual solvent prior to transfer to the DSC pan. The DSC

thermograms were all acquired on a Q25 differential scanning calorimeter (TA Instruments) using a heating and cooling rate of 10°C/min.

Surface Tension Characterizations

Contact angle measurements of each neat film were carried out using water and ethylene glycol by drop shape analysis (Kruss, Germany). The surface tension of each film was calculated by Wu's model:

$$\gamma_{\text{water}}(1 + \cos\theta_{\text{water}}) = \frac{4\gamma_{\text{water}}^d\gamma^d}{\gamma_{\text{water}}^d + \gamma^d} + \frac{4\gamma_{\text{water}}^p\gamma^p}{\gamma_{\text{water}}^p + \gamma^p}$$

$$\gamma_{\text{EG}}(1 + \cos\theta_{\text{EG}}) = \frac{4\gamma_{\text{EG}}^d\gamma^d}{\gamma_{\text{EG}}^d + \gamma^d} + \frac{4\gamma_{\text{EG}}^p\gamma^p}{\gamma_{\text{EG}}^p + \gamma^p}$$

$$\gamma = \gamma^d + \gamma^p$$

where θ is the contact angle of each thin film, and γ is the surface tension of samples, which is equal to the sum of the dispersion (γ^d) and polarity (γ^p) components; γ_{water} and γ_{EG} are the surface tensions of the water and ethylene glycol; and γ_{water}^d , γ_{EG}^d , γ_{water}^p , and γ_{EG}^p are the dispersion and polarity components of γ_{water} and γ_{EG} .

Device Fabrication and Characterizations

All devices were fabricated using chloroform as the solvent with an architecture of glass-ITO-PEDOT:PSS (30 nm)-PDCBT-Cl:acceptor-PDINO-Al (100 nm). The ITO glass substrates were cleaned ultrasonically with detergent, deionized water, acetone and isopropanol for 10 min each and then dried by nitrogen flow. The ITO substrates were then treated with UV ozone for 25 min. And then a PEDOT:PSS (Baytron PVP A14083) layer was spin-coated on the surface followed by baking at 150°C for 15 min. Active layer solutions (weight ratio 1:1) were prepared in CF with a total concentration of 18 mg/mL. Subsequently, the blend solutions were spin-coated at 1,500 rpm to form a photoactive layer with a thickness of about 110 nm measured by Dektak150 profilometer. After that, every thin film was separately placed in a watch glass with 700 μL CS_2 for 2 min. The PDINO in methanol solution was deposited on the top of the active layers. Finally, the aluminum electrode was deposited on the interfacial layer. PM6:Y6 devices were reproduced in our lab according to a previous report.⁵ The active area of the device is 4 mm². J-V curves were measured by Keithley 2400 source meter under 100 mW cm⁻² AM 1.5G simulated solar light illumination provided by an AAA solar simulator (SS-F5-3A, Enli Technology Co. Ltd, Taiwan) calibrated with a standard photovoltaic cell equipped with a KG5 filter in a glove box. The electron mobilities of blend films were conducted by the space-charge-limited-current (SCLC) method. A structure of glass-ZnO (30 nm)-PDCBT-Cl:acceptor-Al (100 nm) was used to construct electron-only devices and the acceptor weight percentage was increased from 0% to 50%. The devices were measured using Keithley 2400 source meter in a glove box under dark. The electron mobilities were calculated by fitting the dark current using the following equation:

$$J = \frac{9}{8}\epsilon_0\epsilon_r\mu\frac{V^2}{L^3}\exp\left(0.89\beta\sqrt{\frac{V}{L}}\right)$$

where J is the current density, L is the film thickness of the active layer, μ is the electron mobility, ϵ_r is the relative dielectric constant of the transport medium, ϵ_0 is the permittivity of free space (8.85×10^{-12} F m⁻¹), $V (= V_{\text{appl}} - V_{\text{bi}})$ is the internal voltage in the device, where V_{appl} is the applied voltage to the device and V_{bi} is the built-in voltage due to the relative work function difference of the two electrodes.

GIWAXS Characterizations

2D-GIWAXS experiments were carried out at Advanced Light Source (ALS), Lawrence Berkeley National Laboratory, on beamline 7.3.3 with an X-ray beam of ~ 10 keV.²⁷ The incidence angle was set to be 0.13° and the sample-to-detector distance was about 277 nm, which was determined by the diffraction peaks of AgB. GIWAXS characterizations were also carried out at the beamline 14B1 of SSRF. An incidence angle of $\sim 0.16^\circ$ was chosen to allow for complete penetration of X-ray into the films. The beam center and sample-to-detector distance were calibrated with LaB₆. The X-ray wavelength was 0.124 nm. The samples were prepared on Si substrates with the same conditions as the device preparation.

RSoXS Characterizations

RSoXS transmission measurements were conducted at beamline 11.0.1.2 at the ALS, Lawrence Berkeley National Laboratory. Details of the experiment setup and data analysis can be found at length as detailed by Gann et al.³⁰ The samples were prepared on PSS-modified glass substrates with the same conditions as the device fabrication. The films were then transferred to 1.5×1.5 mm, 100 nm thick Si₃N₄ membranes supported by a 5×5 mm, 200 mm thick Si frames (Norcada) on water.

Simulation of χ - ϕ Phase Diagrams

The binodal curves presented in this study were simulated by inputting the actual molecular weights and mass densities of PDCBT-Cl and acceptors, with a Flory-Huggins code written by Prof. E. Gomez and J. Litofsky (Penn State).

SUPPLEMENTAL INFORMATION

Supplemental Information can be found online at <https://doi.org/10.1016/j.joule.2020.04.014>.

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AUTHOR CONTRIBUTIONS

L.Y. conceived the idea and designed the experiments; Z.L. performed the DSC and contact angle measurements and tested the devices and stability; Z.L. and Q.W. synthesized the polymer PDCBT-Cl and fabricated the devices; Y.Q. and S.J.S. performed the GIWAXS and RSoXS measurements under the supervision of H.A.; Z.P. acquired part of the GIWAXS data at SSRF; Y.D. assisted with materials synthesis; L.Y., M.L., and Z.L. analyzed all the data and wrote the manuscript with input from H.A. and Y.G. All authors provided comments on the manuscript and contributed to the editing. LY. and M.L. directed the research.

DECLARATIONS OF INTERESTS

The authors declare no competing financial interest.

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REFERENCES

- Yan, C., Barlow, S., Wang, Z., Yan, H., Jen, A.K.Y., Marder, S.R., and Zhan, X. (2018). Non-fullerene acceptors for organic solar cells. *Nat. Rev. Mater.* 3, 18003.
- Liu, Q., Jiang, Y., Jin, K., Qin, J., Xu, J., Li, W., et al. (2020). 18% Efficiency organic solar cells. *Sci. Bull.* 65, 272–275.
- Cui, Y., Han, Y., Hong, L., Zhang, T., Tang, Y., Lin, B., Xian, K., Gao, B., An, C., Bi, P., et al. (2019). 17% efficiency organic photovoltaic cell with superior processability. *Natl. Sci. Rev.* <https://doi.org/10.1093/nsr/nwz200>.
- Jiang, K., Wei, Q., Lai, J.Y.L., Peng, Z., Kim, H.K., Yuan, J., Ye, L., Ade, H., Zou, Y., and Yan, H. (2019). Alkyl chain tuning of small molecule acceptors for efficient organic solar cells. *Joule* 3, 3020–3033.
- Yuan, J., Zhang, Y., Zhou, L., Zhang, G., Yip, H.-L., Lau, T.K., Lu, X., Zhu, C., Peng, H., Johnson, P.A., et al. (2019). Single-junction organic solar cell with over 15% efficiency using fused-ring acceptor with electron-deficient core. *Joule* 3, 1140–1151.
- Li, X., Pan, F., Sun, C., Zhang, M., Wang, Z., Du, J., et al. (2019). Simplified synthetic routes for low cost and high photovoltaic performance n-type organic semiconductor acceptors. *Nat. Commun.* 10, 519.
- Kim, Y., Cook, S., Tuladhar, S.M., Choulis, S.A., Nelson, J., Durrant, J.R., Bradley, D.D.C., Giles, M., McCulloch, I., Ha, C.-S., and Ree, M. (2006). A strong regioregularity effect in self-organizing conjugated polymer films and high-efficiency polythiophene:fullerene solar cells. *Nat. Mater.* 5, 197–203.
- Li, G., Shrotriya, V., Huang, J., Yao, Y., Moriarty, T., Emery, K., and Yang, Y. (2005). High-efficiency solution processable polymer photovoltaic cells by self-organization of polymer blends. *Nat. Mater.* 4, 864–868.
- Holliday, S., Ashraf, R.S., Wadsworth, A., Baran, D., Yousaf, S.A., Nielsen, C.B., Tan, C.H., Dimitrov, S.D., Shang, Z., Gasparini, N., et al. (2016). High-efficiency and air-stable P3HT-based polymer solar cells with a new non-fullerene acceptor. *Nat. Commun.* 7, 11585.
- Guo, X., Cui, C., Zhang, M., Huo, L., Huang, Y., Hou, J., and Li, Y. (2012). High efficiency polymer solar cells based on poly(3-hexylthiophene)/indene-C70 bisadduct with solvent additive. *Energy Environ. Sci.* 5, 7943–7949.
- Xu, X., Zhang, G., Yu, L., Li, R., and Peng, Q. (2019). P3HT-based polymer solar cells with 8.25% efficiency enabled by a matched molecular acceptor and smart green-solvent processing technology. *Adv. Mater.* 31, 1906045.
- Liang, Q., Jiao, X., Yan, Y., Xie, Z., Lu, G., Liu, J., and Han, Y. (2019). Separating crystallization process of P3HT and O-IDTBR to Construct Highly Crystalline Interpenetrating Network with optimized vertical phase separation. *Adv. Funct. Mater.* 29, 1807591.
- Zhang, M., Guo, X., Yang, Y., Zhang, J., Zhang, Z.-G., and Li, Y. (2011). Downwards tuning the HOMO level of polythiophene by carboxylate substitution for high open-circuit-voltage polymer solar cells. *Polym. Chem.* 2, 2900–2906.
- Qin, Y., Uddin, M.A., Chen, Y., Jang, B., Zhao, K., Zheng, Z., Yu, R., Shin, T.J., Woo, H.Y., and Hou, J. (2016). Highly efficient fullerene-free polymer solar cells fabricated with polythiophene derivative. *Adv. Mater.* 28, 9416–9422.
- Wang, Q., Li, M., Zhang, X., Qin, Y., Wang, J., Zhang, J., Hou, J., Janssen, R.A.J., and Geng, Y. (2019). Carboxylate-substituted polythiophenes for efficient fullerene-free polymer solar cells: the effect of chlorination on their properties. *Macromolecules* 52, 4464–4474.
- Jia, X., Liu, G., Chen, S., Li, Z., Wang, Z., Yin, Q., Yip, H.-L., Yang, C., Duan, C., Huang, F., and Cao, Y. (2019). Backbone fluorination of polythiophenes improves device performance of non-fullerene polymer solar cells. *ACS Appl. Energy Mater.* 2, 7572–7583.
- Li, W., Ye, L., Li, S., Yao, H., Ade, H., and Hou, J. (2018). A high-efficiency organic solar cell enabled by the strong intramolecular electron push-pull effect of the nonfullerene acceptor. *Adv. Mater.* 30, 1707170.
- Zhang, S., Qin, Y., Zhu, J., and Hou, J. (2018). Over 14% efficiency in polymer solar cells enabled by a chlorinated polymer donor. *Adv. Mater.* 30, 1800868.
- Meng, L., Yi, Y.Q., Wan, X., Zhang, Y., Ke, X., Kan, B., et al. (2019). A tandem organic solar cell with PCE of 14.52% employing subcells with the same polymer donor and two absorption complementary acceptors. *Adv. Mater.* 31, 1804723.
- Zhang, Y., Kan, B., Sun, Y., Wang, Y., Xia, R., Ke, X., et al. (2018). Nonfullerene tandem organic solar cells with high performance of 14.11. *Adv. Mater.* 30, 1707508.
- Jia, X., Chen, Z., Duan, C., Wang, Z., Yin, Q., Huang, F., and Cao, Y. (2019). Polythiophene derivatives compatible with both fullerene and non-fullerene acceptors for polymer solar cells. *J. Mater. Chem. C* 7, 314–323.
- Lin, Y., He, Q., Zhao, F., Huo, L., Mai, J., Lu, X., Su, C.J., Li, T., Wang, J., Zhu, J., et al. (2016). A facile planar fused-ring electron acceptor for as-cast polymer solar cells with 8.71% efficiency. *J. Am. Chem. Soc.* 138, 2973–2976.
- Zhao, W., Li, S., Yao, H., Zhang, S., Zhang, Y., Yang, B., and Hou, J. (2017). Molecular optimization enables over 13% efficiency in organic solar cells. *J. Am. Chem. Soc.* 139, 7148–7151.
- Zhao, F., Dai, S., Wu, Y., Zhang, Q., Wang, J., Jiang, L., Ling, Q., Wei, Z., Ma, W., You, W., et al. (2017). Single-junction binary-blend nonfullerene polymer solar cells with 12.1% efficiency. *Adv. Mater.* 29, 1700144.
- Lin, Y., Wang, J., Zhang, Z.G., Bai, H., Li, Y., Zhu, D., and Zhan, X. (2015). An electron acceptor challenging fullerenes for efficient polymer solar cells. *Adv. Mater.* 27, 1170–1174.
- Zhao, X., Zhao, Y., Ge, Q., Butrouna, K., Diaio, Y., Graham, K.R., and Mei, J. (2016). Complementary semiconducting polymer blends: the influence of conjugation-break spacer length in matrix polymers. *Macromolecules* 49, 2601–2608.
- Hexemer, A., Bras, W., Glossinger, J., Schaible, E., Gann, E., Kirian, R., MacDowell, A., Church, M., Rude, B., and Padmore, H. (2010). A SAXS/WAXS/GISAXS beamline with multilayer monochromator. *J. Phys. Conf. Ser.* 247, 012007.
- Ye, L., Li, S., Liu, X., Zhang, S., Ghasemi, M., Xiong, Y., Hou, J., and Ade, H. (2019). Quenching to the percolation threshold in organic solar cells. *Joule* 3, 443–458.
- Rivnay, J., Noriega, R., Kline, R.J., Salleo, A., and Toney, M.F. (2011). Quantitative analysis of lattice disorder and crystallite size in organic semiconductor thin films. *Phys. Rev. B* 84, 045203.
- Gann, E., Young, A.T., Collins, B.A., Yan, H., Nasiatka, J., Padmore, H.A., Ade, H., Hexemer, A., and Wang, C. (2012). Soft x-ray scattering facility at the Advanced Light Source with real-time data processing and analysis. *Rev. Sci. Instrum.* 83, 045110.
- Jiao, X., Ye, L., and Ade, H. (2017). Quantitative morphology-performance correlations in organic solar cells: insights from soft X-ray scattering. *Adv. Energy Mater.* 7, 1700084.
- Mukherjee, S., Jiao, X., and Ade, H. (2016). Charge creation and recombination in multi-

- length scale polymer:fullerene BHJ solar cell morphologies. *Adv. Energy Mater.* 6, 1600699.
33. Gao, K., Jo, S.B., Shi, X., Nian, L., Zhang, M., Kan, Y., et al. (2019). Over 12% efficiency nonfullerene all-small-molecule organic solar cells with sequentially evolved multilength scale morphologies. *Adv. Mater.* 31, 1807842.
 34. Ye, L., Hu, H., Ghasemi, M., Wang, T., Collins, B.A., Kim, J.H., Jiang, K., Carpenter, J.H., Li, H., Li, Z., et al. (2018). Quantitative relations between interaction parameter, miscibility and function in organic solar cells. *Nat. Mater.* 17, 253–260.
 35. Yi, N., Ai, Q., Zhou, W., Huang, L., Zhang, L., Xing, Z., Li, X., Zeng, J., and Chen, Y. (2019). Miscibility matching and bimolecular crystallization affording high-performance ternary nonfullerene solar cells. *Chem. Mater.* 31, 10211–10224.
 36. Nishi, T., and Wang, T.T. (1975). Melting point depression and kinetic effects of cooling on crystallization in poly(vinylidene fluoride)-poly(methyl methacrylate) mixtures. *Macromolecules* 8, 910–915.
 37. Kozub, D.R., Vakhshouri, K., Orme, L.M., Wang, C., Hexemer, A., and Gomez, E.D. (2011). Polymer crystallization of partially miscible polythiophene/fullerene mixtures controls morphology. *Macromolecules* 44, 5722–5726.
 38. Li, S., Ye, L., Zhao, W., Yan, H., Yang, B., Liu, D., Li, W., Ade, H., and Hou, J. (2018). A wide band gap polymer with a deep highest occupied molecular orbital level enables 14.2% efficiency in polymer solar cells. *J. Am. Chem. Soc.* 140, 7159–7167.
 39. Feng, H., Yi, Y.Q.Q., Ke, X., Yan, J., Zhang, Y., Wan, X., Li, C., Zheng, N., Xie, Z., and Chen, Y. (2019). New anthracene-fused nonfullerene acceptors for high-efficiency organic solar cells: energy level modulations enabling match of donor and acceptor. *Adv. Energy Mater.* 9, 1803541.
 40. Kouijzer, S., Michels, J.J., van den Berg, M., Gevaerts, V.S., Turbiez, M., Wienk, M.M., and Janssen, R.A. (2013). Predicting morphologies of solution processed polymer:fullerene blends. *J. Am. Chem. Soc.* 135, 12057–12067.
 41. Ye, L., Collins, B.A., Jiao, X., Zhao, J., Yan, H., and Ade, H. (2018). Miscibility-function relations in organic solar cells: significance of optimal miscibility in relation to percolation. *Adv. Energy Mater.* 8, 1703058.
 42. Bartelt, J.A., Bailey, Z.M., Hoke, E.T., Mateker, W.R., Douglas, J.D., Collins, B.A., Tumbleston, J.R., Graham, K.R., Amassian, A., Ade, H., et al. (2013). The importance of fullerene percolation in the mixed regions of polymer-fullerene bulk heterojunction solar cells. *Adv. Energy Mater.* 3, 364–374.
 43. Liu, Y., Zhao, J., Li, Z., Mu, C., Ma, W., Hu, H., Jiang, K., Lin, H., Ade, H., and Yan, H. (2014). Aggregation and morphology control enables multiple cases of high-efficiency polymer solar cells. *Nat. Commun.* 5, 5293.
 44. Treat, N.D., Westacott, P., and Stingelin, N. (2015). The power of materials science tools for gaining insights into organic semiconductors. *Annu. Rev. Mater. Res.* 45, 459–490.
 45. Matkar, R.A., and Kyu, T. (2006). Role of crystal-amorphous interaction in phase equilibria of crystal-amorphous polymer blends. *J. Phys. Chem. B* 110, 12728–12732.
 46. Rathi, P., Huang, T.M., Dayal, P., and Kyu, T. (2008). Crystalline-amorphous interaction in relation to the phase diagrams of binary polymer blends containing a crystalline constituent. *J. Phys. Chem. B* 112, 6460–6466.
 47. Ghasemi, M., Hu, H., Peng, Z., Rech, J.J., Angunawela, I., Carpenter, J.H., Stuard, S.J., Wadsworth, A., McCulloch, I., You, W., and Ade, H. (2019). Delineation of thermodynamic and kinetic factors that control stability in non-fullerene organic solar cells. *Joule* 3, 1328–1348.
 48. Vakhshouri, K., Kozub, D.R., Wang, C., Salleo, A., and Gomez, E.D. (2012). Effect of miscibility and percolation on electron transport in amorphous poly(3-hexylthiophene)/phenyl-C61-butyric acid methyl ester blends. *Phys. Rev. Lett.* 108, 026601.
 49. Michels, J.J. (2011). Surface-directed spinodal decomposition of solvent-quenched organic transistor blends. *ChemPhysChem* 12, 342–348.
 50. Vaynzof, Y., Kabra, D., Zhao, L., Chua, L.L., Steiner, U., and Friend, R.H. (2011). Surface-directed spinodal decomposition in Poly[3-hexylthiophene] and C₆₁-butyric acid methyl ester blends. *ACS Nano* 5, 329–336.
 51. Zhang, L., Yi, N., Zhou, W., Yu, Z., Liu, F., and Chen, Y. (2019). Miscibility tuning for optimizing phase separation and vertical distribution toward highly efficient organic solar cells. *Adv. Sci. (Weinh.)* 6, 1900565.
 52. Liu, S., Chen, D., Zhou, W., Yu, Z., Chen, L., Liu, F., and Chen, Y. (2019). Vertical distribution to optimize active layer morphology for efficient all-polymer solar cells by J71 as a compatibilizer. *Macromolecules* 52, 4359–4369.
 53. Li, N., Perea, J.D., Kassari, T., Richter, M., Heumüller, T., Matt, G.J., Hou, Y., Güldal, N.S., Chen, H., Chen, S., et al. (2017). Abnormal strong burn-in degradation of highly efficient polymer solar cells caused by spinodal donor-acceptor demixing. *Nat. Commun.* 8, 14541.
 54. Zhou, R., Jiang, Z., Yang, C., Yu, J., Feng, J., Adil, M.A., Deng, D., Zou, W., Zhang, J., Lu, K., et al. (2019). All-small-molecule organic solar cells with over 14% efficiency by optimizing hierarchical morphologies. *Nat. Commun.* 10, 5393.
 55. Osaka, I., Abe, T., Shinamura, S., and Takimiya, K. (2011). Impact of isomeric structures on transistor performances in naphthodithiophene semiconducting polymers. *J. Am. Chem. Soc.* 133, 6852–6860.
 56. Liang, Z., Li, M., Zhang, X., Wang, Q., Jiang, Y., Tian, H., and Geng, Y. (2018). Near-infrared absorbing non-fullerene acceptors with selenophene as π bridges for efficient organic solar cells. *J. Mater. Chem. A* 6, 8059–8067.
 57. Zhou, Q., Jiang, Y., Du, T., Wang, Z., Liang, Z., Han, Y., Deng, Y., Tian, H., and Geng, Y. (2019). Diketopyrrolopyrrole-based small molecules for solution-processed n-channel organic thin film transistors. *J. Mater. Chem. C* 7, 13939–13946.
 58. Wu, Y., Zheng, Y., Yang, H., Sun, C., Dong, Y., Cui, C., Yan, H., and Li, Y. (2020). Rationally pairing photoactive materials for high-performance polymer solar cells with efficiency of 16.53%. *Sci. China Chem.* 63, 265–271.
 59. Xiong, J., Jin, K., Jiang, Y., Qin, J., Wang, T., Liu, J., et al. (2019). Thiolactone copolymer donor gifts organic solar cells a 16.72% efficiency. *Sci. Bull.* 64, 1573–1576.
 60. Fan, B., Zhang, D., Li, M., Zhong, W., Zeng, Z., Ying, L., Huang, F., and Cao, Y. (2019). Achieving over 16% efficiency for single-junction organic solar cells. *Sci. China Chem.* 62, 746–752.
 61. Wang, T., Qin, J., Xiao, Z., Meng, X., Zuo, C., Yang, B., et al. (2020). A 2.16 eV bandgap polymer donor gives 16% power conversion efficiency. *Sci. Bull.* 65, 179–181.
 62. Fan, B., Du, X., Liu, F., Zhong, W., Ying, L., Xie, R., Tang, X., An, K., Xin, J., Li, N., et al. (2018). Fine-tuning of the chemical structure of photoactive materials for highly efficient organic photovoltaics. *Nat. Energy* 3, 1051–1058.
 63. Zhang, B., Yu, Y., Zhou, J., Wang, Z., Tang, H., Xie, S., Xie, Z., Hu, L., Yip, H.L., Ye, L., et al. (2020). 3,4-Dicyanophenylene—a versatile building block for efficient nonfullerene polymer solar cells. *Adv. Energy Mater.* 10, 1904247.
 64. Ye, L., Xiong, Y., Chen, Z., Zhang, Q., Fei, Z., Henry, R., et al. (2019). Sequential deposition of organic films with eco-compatible solvents improves performance and enables over 12%-efficiency nonfullerene solar cells. *Adv. Mater.* 31, 1808153.
 65. Zhu, Y., Gadisa, A., Peng, Z., Ghasemi, M., Ye, L., Xu, Z., Zhao, S., and Ade, H. (2019). Rational strategy to stabilize an unstable high-efficiency binary nonfullerene organic solar cells with a third component. *Adv. Energy Mater.* 9, 1900376.
 66. Hu, H., Li, Y., Zhang, J., Peng, Z., Ma, L.-k., Xin, J., Huang, J., Ma, T., Jiang, K., Zhang, G., et al. (2018). Effect of ring-fusion on miscibility and domain purity: key factors determining the performance of PDI-based nonfullerene organic solar cells. *Adv. Energy Mater.* 8, 1800234.
 67. Trafton, A. (2020). Artificial intelligence yields new antibiotic (a deep-learning model identifies a powerful new drug that can kill many species of antibiotic-resistant bacteria). MIT News Office, February 20, 2020. <http://news.mit.edu/2020/artificial-intelligence-identifies-new-antibiotic-0220>.